

CSE141 INTRODUCTION TO PROGRAMMING

Final Examination

Max Marks: 50

Time Allowed: 3 hours

Answer the questions in the spaces provided on the question sheets.
Please give clear and rigorous answers. Be to the point. Show your work.

Name: _____

ERP: _____

Question:	True/False	MCQ's	Trace/Debug	Methods	Arrays	Recursion	Classes	Total
Marks:	10	9	11	5	5	5	5	50
Score:								

Question 1: True/False 10 marks

True/False questions. Circle the correct answer.

- (a) **T F** A single `System.out.println` statement in a Java program could result in multiple lines of output.
- (b) **T F** Multiple `System.out.println` statements in a Java program could result in a single line of output.
- (c) **T F** Variable names in Java must have at least one upper-case letter.
- (d) **T F** To end a Java program both the `break` or the `exit` statement may be used.
- (e) **T F** An if-else statement could be rewritten using two if statements without an else.
- (f) **T F** The following statements give a run-time error in Java:

```
String name = "123";  
if ( name.length() == 3) {  
    System.out.println(name.charAt( 3 ));  
}
```

- (g) **T F** The output of the following statements is: 3 Done

```
int x = 2;  
int y = 1;  
System.out.print( " " + x + y );  
System.out.println( " Done " );
```

- (h) **T F** To check and see if the value of variable x is between 3 and 9 we could use:

```
int x = 4;  
if ( 3 < x < 9 ) {  
    System.out.println( "x is between 3 and 9" );  
}
```

- (i) **T F** The output of the following lines of code is: One Done

```
int x = 2;
if( x = 1);
    System.out.print("One ");
System.out.println("Done");
```

- (j) **T F** The output of the statement below is: 1.7

```
int x = 5;
int y = 3;
double answer = 0.0;
answer = x / y;
System.out.printf("%2.1f", answer);
```

Question 2: MCQ's 9 marks

Circle the correct answer.

- (a) [2 marks] Consider the class given below, along with a client for it.

```
class ClassA {
    private int x;
    public ClassA() {
        x = 1;
    }
    public void addValue(int val) {
        x = x + val;
    }
} //end ClassA

class ClassAClient {
    public void doIt() {
        int value = 7;
        ClassA instance1 = new ClassA();
        instance1.addValue( 5);
        System.out.println("value is: " + instance1.x);
    }
} //end ClassAClient
```

When running method doIt () in the ClassAClient class, the output will be:

- i. value is: 5
- ii. value is: 6
- iii. value is: 7
- iv. doesn't compile
- v. None of the above

- (b) [1½ marks] Consider the program segment given below. (Remember that the % operator is the modulus operator, which gives the remainder after integer division.) Its output is:

```
int i=1;
while (i<=30) {
    i++;
    System.out.print(i);
    if( i%2 == 0)
        System.out.println();
    i++;
}
```

- i. All the numbers from 1 to 30, with a line break after the number 10
 - ii. Even numbers from 1 to 30 all one one line
 - iii. Odd numbers from 1 to 30, each on its own line
 - iv. Even numbers from 1 to 30, each on its own line
 - v. None of the above
- (c) [2 marks] Consider method `second` shown below, which itself uses method `first`. For positive numbers, how would you best describe its return value?

```
public int first(int x, int y) {
    int z=0;
    for (int i=0; i<y; i++) {
        z += x;
    }
    return z;
}

public int second(int x, int y) {
    int z=1;
    for (int i=0; i<y; i++) {
        z = first( z, x);
    }
    return z;
}
```

- i. $x + y$
- ii. $x * x$
- iii. $x * y$
- iv. x^y
- v. None of the above

(d) [1½ marks] The output of the following code in Java is equal to:

```
int value = 2;
int limit = 5;
int result = 1;
for(int x=0; x<limit; x++) {
    result = result * value;
}
System.out.println( result);
```

- i. $2 * 5$
- ii. $2 + 5$
- iii. 25
- iv. 32
- v. None of the above

(e) [1 mark] If we start with the array 6 1 14 10 5 12 11 9, which sorting algorithms will encounter 1 5 6 10 14 12 11 9 during the sorting process?

- i. Insertion sort
- ii. Mergesort
- iii. All of the above
- iv. None of the above

(f) [1 mark] In worst-case, the number of steps taken by *binary search* when applied on an array of size N is proportional to

- i. constant
- ii. $\log N$
- iii. N
- iv. $N \log N$
- v. N^2

Question 3: Trace/Debug 11 marks

- (a) [2 marks] Point out any errors in the following statements.

Statement	Error (if any)
float = 3.14*2;	
float stu marks = 40.5;	
FirstName = 'Donald';	
char n = '7';	

- (b) [2 marks] Consider the following method:

```

public static void mystery(int n) {
    if (n > 100) {
        System.out.print(n);
    } else {
        mystery(2 * n);
        System.out.print(", " + n);
    }
}

```

For each of the following calls, indicate the output that is produced by the method:

i. `mystery(60);`

ii. `mystery(41);`

.....

- (c) [1½ marks] What is the output of the following :

```

public static void main(String[] args) {
    int result = mystery2(348);
    System.out.printf("Result is %d : ", result);
}

public static int mystery2(int num) {
    if (num < 10)
        return num;
    else
        return num%10 + mystery2(num/10);
}

```

.....

- (d) [2 marks] Consider the following method, `mystery3`:

```
public static void mystery3(int[] a, int[] b) {  
    for (int i = 0; i < a.length; i++) {  
        a[i] += b[b.length - 1 - i];  
    }  
}
```

What are the values of the elements in array `x` after the following code executes?

```
int[] x = {1, 3, 5, 7, 9};  
int[] y = {1, 4, 9, 16, 25};  
mystery3(x, y);
```

.....
.....

- (e) [1 mark] What does the following code fragment print?

```
String string1 = "hello";  
String string2 = string1;  
string1 = "world";  
System.out.println(string1);  
System.out.println(string2);
```

.....
.....

- (f) [1 mark] What does the following code fragment print?

```
String s = "Hello World";  
s.toUpperCase();  
s.substring(6, 11);  
System.out.println(s);
```

.....
.....

- (g) [1½ marks] What does the following recursive function return?

```
public static String mystery4(String s) {  
    int N = s.length();  
    if (N <= 1) return s;  
    String a = s.substring(0, N/2);  
    String b = s.substring(N/2, N);  
    return mystery4(b) + mystery4(a);  
}
```

.....
.....

Question 4: Methods 5 marks

- (a) [$2\frac{1}{2}$ marks] Write a static method called `isSorted` that given an array of integers returns **true** if array is sorted in ascending order and **false** otherwise.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) [$2\frac{1}{2}$ marks] Write a static method called `sumTo` that accepts an integer parameter n and returns a real number representing the sum of the first n reciprocals. In other words, `sumTo( $n$ )` returns $(1 + 1/2 + 1/3 + 1/4 + \dots + 1/n)$. For example, `sumTo(2)` should return `1.5`. The method should return `0.0` if it is passed the value ≤ 0 .

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Write a method called `contains` that accepts two arrays of integers `A` and `B` as parameters and that returns a boolean value indicating whether or not the sequence of elements in `B` appears in `A` (**true** for yes, **false** for no). The sequence must appear consecutively and in the same order. For example, consider the following arrays:

The call of `contains(list1, list2)` should return **true** because the sequence of values in `list2{1, 2, 1}` is contained in `list1` starting at index 5. If `list2` had stored the values `{2, 1, 2}`, the call of `contains(list1, list2)` would return **false**. Any two lists with identical elements are considered to contain each other. Every array contains the empty array, and the empty array does not contain any arrays other than the empty array itself.

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

Write a *recursive* method called `doubleDigits` that accepts an integer n as a parameter and returns the integer obtained by replacing every digit of n with two of that digit. For example, `doubleDigits(348)` should return 334488. The call `doubleDigits(x)` for $x \leq 0$ should return 0.

Design a class named `Rectangle` to represent a rectangle. The class contains:

- Two double data fields named `width` and `height` that specify the width and height of the rectangle. The default values are 1 for both `width` and `height`.
- A no-arg constructor that creates a default rectangle.
- A constructor that creates a rectangle with the specified `width` and `height`.
- A method named `getArea()` that returns the area of this rectangle.

[illegible]